

GaugeAnything: Promptable Quantitative Inspection for Industrial Micro-Vision

Masks In, Millimeters Out

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github.com/falcons-eyes/GaugeAnything

Abstract

Vision foundation models stop at perception: Segment Anything yields masks, Depth Anything relative depth, counting models counts—but industrial inspection decisions are made in *physical units*: “crack width $0.42\text{ mm} \pm 0.05$, condition Fair.” As of mid-2026 no promptable model emits such measurements, and—strikingly—no public dataset pairs defect photos with physically measured widths. We introduce GaugeAnything, a promptable quantitative inspection pipeline that couples a concept-promptable segmentation backbone (SAM 3) with a *metrology core*: tilt-robust pixel-to-millimeter scale resolvers, mask-geometry measurement, a boundary-regime router (sharp/fuzzy/boundaryless), and a signal-based width estimator. Our central finding is a decomposition we call “**mask for where, signal for how wide**”: reading width from binary mask geometry costs $4\text{--}6\times$ accuracy, whereas using the mask only to localize the centerline and regressing width from the raw intensity profile attains **$23.2\text{ }\mu\text{m}$ median error** against manually measured physical ground truth (krkCMd, 19,098 profiles)—within $2\times$ of the dataset authors’ specialized supervised model. On mm-accurate industrial CAD ground truth (T-LESS) the full promptable pipeline measures part dimensions at **2.5% median error**, statistically indistinguishable from a perfect-mask ceiling of 2.83% . We further report the first peer-comparable SAM 3 crack-segmentation IoU (0.442 ± 0.011 crack-only, $2.44\times$ classical baselines), a calibration ladder in which a six-parameter logit-threshold + quantile scheme (0.437 rel. err.) beats a 1.9M-parameter neural refiner (0.564), and honest negative results: zero-shot counting collapses on dense touching parts regardless of prompt wording, and an alpha-matting head that wins $20\times$ on synthetic data initially failed to transfer until the synthesis was made directionally realistic (real-fray IoU $0.483 \rightarrow 0.949$). All code, audited benchmarks, and five task heads are released.

1 Introduction

Industrial inspection is a measurement discipline. Whether a concrete crack requires repair depends on whether its width exceeds a codified threshold (e.g. 0.3 mm); whether an assembly passes depends on bolt counts and spacings; maintenance scheduling depends on graded severity. Yet the modern “promptable anything” toolbox—Segment Anything [1–3], Depth Anything [4], class-agnostic counting [5]—stops one step short of the quantity that drives the decision. The model that tells an inspector *how many millimeters* does not exist, and the gap is structural: our survey (§2) finds that the closest prior work validates promptable measurement on a single object at “ $\pm 10\%$ ” [6], and that public crack datasets provide pixel masks only—none pairs images with physically measured widths.

GaugeAnything fills this gap with a deliberately decomposed design (§3): a frozen concept-promptable backbone (SAM 3) proposes instances from a noun-phrase prompt; a *metrology core*

converts instances into *Inspection Atoms* {mask, class, count, size in mm $\pm\sigma$, condition grade, confidence}. The core contains (i) *scale resolvers*—fiducial markers, known-dimension objects, and a tilt-robust homography (*PlaneScale*) that reduces a 19.3% scale error at 50° camera tilt to 0.7%; (ii) *mask geometry*—skeleton + Euclidean distance transform width profiles, equivalent diameters, spacings; (iii) a *boundary-regime router* that dispatches sharp-boundary defects to binary segmentation, fuzzy boundaries (fray) to alpha matting, and boundaryless fields (uneven/mura) to illumination-residual modeling; and (iv) a *signal-based width estimator* that is the paper’s central contribution.

Central finding: mask for *where*, signal for *how wide*. A binary mask quantizes a gradual intensity transition into a hard boundary, destroying sub-pixel evidence. We show this is the dominant error source for thin-structure measurement: on scanner imagery with manually measured physical widths (krkCMd [7]), widths read from SAM 3 mask geometry err by 144–186 μm , while a 1-D CNN regressing width from the raw 501-pixel brightness profile—using the mask *only* to localize the profile position—achieves 39.9 μm MAE and 23.2 μm median where localization succeeds (§4.6). The information was in the image all along; the binary mask was discarding it.

Contributions.

1. **Task & system:** Promptable Quantitative Inspection—image + noun phrase $\rightarrow$ Inspection Atoms—implemented as an open, audited pipeline on a frozen SAM 3 backbone with a license-clean metrology core.
2. **Signal-based metrology:** the *where/how-wide* decomposition, validated on physical ground truth: 23.2 μm median (localization-gated) vs. 144–186 μm for mask geometry; on T-LESS CAD ground truth, promptable part measurement at 2.5% median $\approx$ the 2.83% perfect-mask ceiling.
3. **Calibration ladder:** a systematic, honestly-reported progression for width bias—raw 0.730 $\rightarrow$ neural refiner 0.564 $\rightarrow$ 5-number quantile calibration 0.480 $\rightarrow$ logit-threshold + quantile 0.437 (held-out sources)—in which the simple method beats the learned one, and we say so.
4. **First numbers:** the first peer-comparable SAM 3 crack IoU (0.442 ± 0.011 crack-only; the field’s supervised SOTA reports two-class mIoU, a metric trap we quantify); the first promptable physical crack-width MAE; per-class soft-map results on Magnetic-Tile fray/uneven, which prior work reports only in aggregate.
5. **Honest negatives, released:** prompt-synonym collapse and its ensemble rescue; zero-shot dense counting failure (six prompts, 0% acc@10%) with a partial SAHI recovery; a synthetic-to-real matting failure and its directional-synthesis fix; an equivalent-width hypothesis rejected by textured backgrounds.

2 Related Work

Promptable segmentation. SAM [1], SAM 2 [2] and SAM 3 [3] established promptable mask generation; SAM 3 adds concept prompts (noun phrases) with cgF1 54.1 on SA-Co/Gold and strong counting (CountBench MAE 0.12). Its authors note weak generalization to “fine-grained out-of-domain concepts in niche visual domains”—precisely the industrial regime we probe. No peer-reviewed SAM 3 crack-segmentation pixel IoU exists prior to this work.

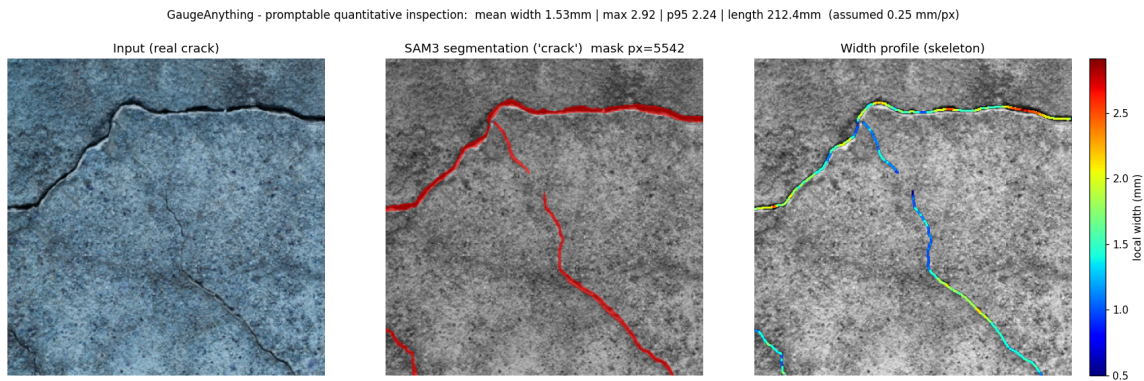


Figure 1: GaugeAnything on a real concrete surface: prompt “crack” → SAM 3 instances → skeleton+EDT width profile → physical width (assumed scale shown; absolute scale comes from markers or known-dimension objects).

Crack segmentation. Supervised SOTA on CrackSeg9k [8] reports *two-class* mIoU (background + crack): DeepLabV3+ 0.7599, HrSegNet-B48 80.32 [9], CrackMamba 81.75 [10]; because background IoU is ~ 0.95 , these are not comparable to crack-only IoU. SAM-adaptation work reports crack-only IoU: CrackSAM (LoRA) 0.6416 in-domain [11], SAC (LayerNorm-only tuning) 44.13 after training on OmniCrack30k [12]; OmniCrack30k [13] finds nnU-Net beats all specialized crack networks (cIoU_{4px} 64%). Our zero-shot 0.442 equals SAC’s *fine-tuned* crack IoU.

Crack width measurement. RGB-only skeleton/orthogonal methods report $r=0.962$, RMSE 0.24 mm on synthetic cracks [14]; depth-assisted systems reach < 0.05 mm with a LiDAR camera [15]; laser-projection systems 0.02–0.57 mm [16]. krkCmd [7] releases 19,098 cross-crack brightness profiles with manual widths—to our knowledge the only public physical width GT—whose authors’ deep model (DLM) attains 11.1 μ m MAE. We position GaugeAnything as RGB-only, single-image, and promptable, between the RGB-only and sensor-assisted regimes.

Promptable measurement. Measure Anything [6] (SAM-2 + stereo) reports stem-diameter automation and a single “ $\pm 10\%$ ” bottle validation, with no caliper MAE; agricultural stereo systems are similar. A quantitative, physically grounded promptable-metrology table has been empty; we contribute its first entries.

Counting and anomaly detection. FSC-147 SOTA: CountGD test MAE 6.75 [5], GeCo 7.91 [17]; supervised rebar counting reaches count-accuracy 86.27% [18]. Zero-shot text-prompted anomaly detection (AnomalyCLIP [19], pixel-AUROC 91.1 on MVTec) localizes but does not measure. On Magnetic-Tile defects [20], supervised mIoU reaches 86.5 [21]; per-class fray/uneven numbers are rarely tabulated.

3 Method

3.1 Task: Promptable Quantitative Inspection

Given an image I , a noun-phrase prompt p (e.g. “crack”, “hex bolt”), and an optional scale reference (fiducial marker size or a known-dimension object class), GaugeAnything returns a set of *Inspection*

Atoms: $a_i = (m_i, c_i, s_i, g_i, u_i)$ —instance mask, class, size measurements in physical units, condition grade, and confidence/uncertainty. Counting is $|\{a_i\}|$; spacing is pairwise centroid distance in mm.

3.2 Backbone and prompt robustness

We use SAM 3 frozen, via its image concept-segmentation interface. Because concept prompts are brittle—“fracture” and “pit” score 0.000 where “crack” and “hole” score 0.374/0.352—we wrap the backbone in a *prompt-set ensemble*: a curated synonym set per defect family, instances unioned with IoU-0.5 deduplication keeping the highest-scoring instance. The raw model also exposes per-query soft mask logits (range $[-67, 10]$ pre-sigmoid), which we exploit in §4.3 and for uncertainty.

3.3 Metrology core

Scale. Pixel-to-mm conversion comes from (i) ArUco/ChArUco fiducials; (ii) known-dimension objects (bolt-head across-flats, coin diameters); (iii) *PlaneScale*: the four marker corners define a homography H to the metric plane, and all measurements are taken in plane coordinates with a local scale at each instance centroid. At 50° tilt, naive single-factor scaling errs 19.3%; PlaneScale errs 0.7% (§4.8). **Geometry.** Thin structures: skeletonization + Euclidean distance transform; width $= 2 \cdot \text{EDT}$ at skeleton points, yielding a width profile (mean, P95, max). Blobs: equivalent diameter $\sqrt{4A/\pi}$. The thin/blob choice is automatic via elongation. **Regime router.** Defects differ in boundary physics: sharp (crack on smooth concrete), fuzzy (fray—frayed bristle-like boundaries), boundaryless (uneven/mura—a gradual field). A statistics-based router dispatches to binary masks, alpha matting (trained on directional synthetic compositing), or illumination-residual modeling (2-D polynomial fit + ISO-25178-style roughness), respectively. Soft outputs are measured by $\text{area} = \sum \alpha$ with uncertainty $\text{Var} \approx \sum \alpha(1 - \alpha)$.

3.4 Signal-based width: mask for where, signal for how wide

For thin-structure width we *do not* read the mask boundary. The mask (tiled SAM 3 inference; darkest long connected component; skeletonization) provides only the centerline. At each measurement station we extract the raw image intensity profile perpendicular to the centerline (snap-to-valley re-centering, ± 80 px), and a 1-D CNN (three conv blocks, ~ 0.2 M parameters), trained on krkCmd’s 14,424 training profiles (CC BY 4.0), regresses physical width directly from the 501-sample profile. A correlation-based confidence gate rejects stations whose profile cannot be trusted; gated stations are reported as *not measurable* rather than guessed—we consider this gating part of the instrument’s contract, and we report coverage alongside accuracy.

4 Experiments

All numbers were produced on a single NVIDIA GB10 (aarch64); protocols enforce val/test separation (configuration selected on validation only), held-out test *sources*, multi-seed reporting where applicable, and crack-only metrics with empty-GT images scored separately. Datasets and licenses are catalogued in the repository ([paper/DATASETS.md](#)); deprecated pre-audit numbers are retained in the results log.

4.1 Zero-shot promptable segmentation (Gauge-Bench)

On CrackSeg9k (crack-only IoU, empty-GT excluded, 3 seeds): frangi 0.115 ± 0.005 , adaptive threshold 0.181 ± 0.006 , **SAM 3 zero-shot** 0.442 ± 0.011 ($2.44\times$ the best classical), with a non-

Table 1: Calibration ladder for crack width on held-out sources ($n=219$). Lower is better. “Params” counts learned/selected quantities.

Method	Params	Rel. err. ↓	Bias
Raw SAM 3 mask width ($\theta=0.5$)	0	0.730	+0.680
Global affine calibration	2	0.679	+0.633
M2 neural refiner (UNet)	1.9M	0.564	+0.503
Quantile-ratio calibration	5	0.480	+0.411
Logit iso-level $\theta^*=0.7$	1	0.493	+0.406
$\theta^* + \text{quantile}$	6	0.437	+0.367

crack clean rate of 0.68 vs. 0.00 for the adaptive baseline. We emphasize the *metric trap*: supervised crack SOTA reports two-class mIoU (0.77–0.82), which is not comparable; in crack-only terms our zero-shot equals SAC’s fine-tuned 44.13 [12]. Cross-domain, the same prompt-driven pipeline transfers from concrete to magnetic tile (crack 0.454; blowhole via “hole” 0.429 with automatic blob→diameter dispatch).

4.2 Segmentation rank $\neq$ measurement rank

On width fidelity (GT 11.3 px), the best-IoU method is not the best measure: adaptive (IoU 0.181) errs 43.5% relative, SAM 3 (IoU 0.442) errs 62.9%. This dissociation motivates everything that follows.

4.3 The calibration ladder

With three crack sources held out entirely (cfd, cracktree200, deepcrack; $n=219$), Table 1 traces width relative error. The 1.9M-parameter measurement-aware neural refiner (M2) improves raw SAM 3 but *loses* to a five-number quantile calibration fit on training sources; exploiting the exposed mask logits—selecting the binarization iso-level $\theta^*=0.7$ on train/val—and stacking quantile calibration reaches 0.437 with zero learned parameters. Width bias is domain-dependent (train sources -0.17 , held-out $+0.68$), so a single global correction cannot finish the job: the residual is a property of the mask, which is what §4.6 removes.

4.4 Promptable part metrology on CAD ground truth (T-LESS)

BOP-format CAD models are metrically exact: projecting model vertices through the annotated pose yields, for the maximal projected chord, an exact mm distance. With *perfect* (ground-truth) visible masks and a single pose-depth plane scale, measurement attains median 2.83% (94% within 10%; worst-case 34.6% on depth-elongated objects)—a methodological ceiling for single-depth plane scaling on 3-D parts. Replacing perfect masks with SAM 3 zero-shot (concrete-noun prompts: “electrical component”, “plastic part”, “white object”): match rate 100%, mask IoU 0.94, and **measurement median** 2.5%—the segmentation step costs essentially nothing. The abstract prompt “industrial part” matches 0%: synonym collapse reappears in a third domain, reinforcing the prompt-ensemble design.

4.5 Real-photo consistency (coins)

On 22 real photos (8–60 same-denomination euro coins each), leave-one-out known-object scaling (each coin measured using the others as reference) yields mean error 1.74%, 100% within 5%. This

E-mm-1: real-photo measurement consistency (no marker, no field rig)

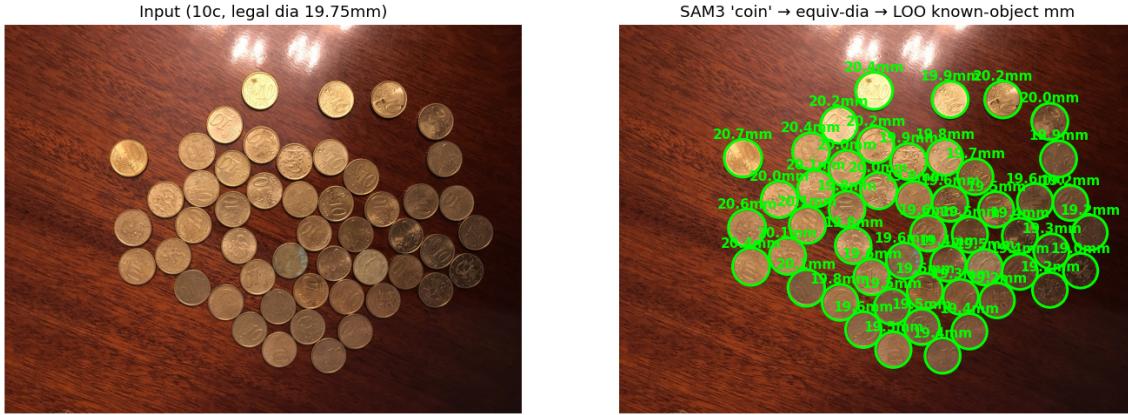


Figure 2: Real-photo known-object consistency (E-mm-1): each coin is measured using the other same-denomination coins as scale reference; mean leave-one-out error 1.74% over 22 scenes.

Table 2: Physical crack-width error on krkCMd (μm). “Gated” = stations passing the localization confidence gate (coverage 46–66%).

Level	Method	MAE
Profile (table)	authors’ DLM (supervised)	11.1
Profile (table)	ours: 1-D CNN	17.9
Profile (table)	valley rule + linear cal. (5-fold)	27.8 ± 2.5
Image, oracle pos.	ours: 1-D CNN	26.2
Image, promptable (gated)	ours: SAM 3 locate + 1-D CNN	39.9 (median 23.2)
Image, promptable	SAM 3 mask geometry (θ^* , best)	144–186

validates the segmentation→diameter chain on real imagery; the absolute marker chain is validated synthetically (ArUco 0.38%; e2e 5.6%).

4.6 Physical crack width: from profiles to the promptable chain

Profile level. On krkCMd’s manual widths (group-split test 4,674 profiles), a deterministic valley rule with linear calibration attains $25.9 \mu\text{m}$ (5-fold 27.8 ± 2.5 ; leave-one-series worst 46.7), matching the authors’ classical AED (26.5); their supervised DLM anchors at $11.1 \mu\text{m}$. **Image level, mask geometry.** Reconstructing every profile’s image position from the released ROI coordinates (correlation-verified, 0.95 gate), widths read from SAM 3 mask geometry err 144–186 μm even after θ^* transfer (which itself improves the default by $\sim 30\%$, showing the logit knob generalizes across datasets and into physical units). **Image level, signal-based (ours).** The 1-D profile CNN reaches $17.9 \mu\text{m}$ on the table test split; applied at *oracle* positions in the image, $26.2 \mu\text{m}$; applied at SAM 3-localized, valley-snapped positions, **39.9 μm MAE / 23.2 μm median** on the 46–66% of stations passing the localization gate (failed stations: 186 μm , reported as not-measurable). Width information survives imaging; binary masks were the bottleneck, and localization coverage—a classical-vision engineering task—is what remains.

E-mm-3 krkCMD: 501-pixel cross-crack brightness profiles

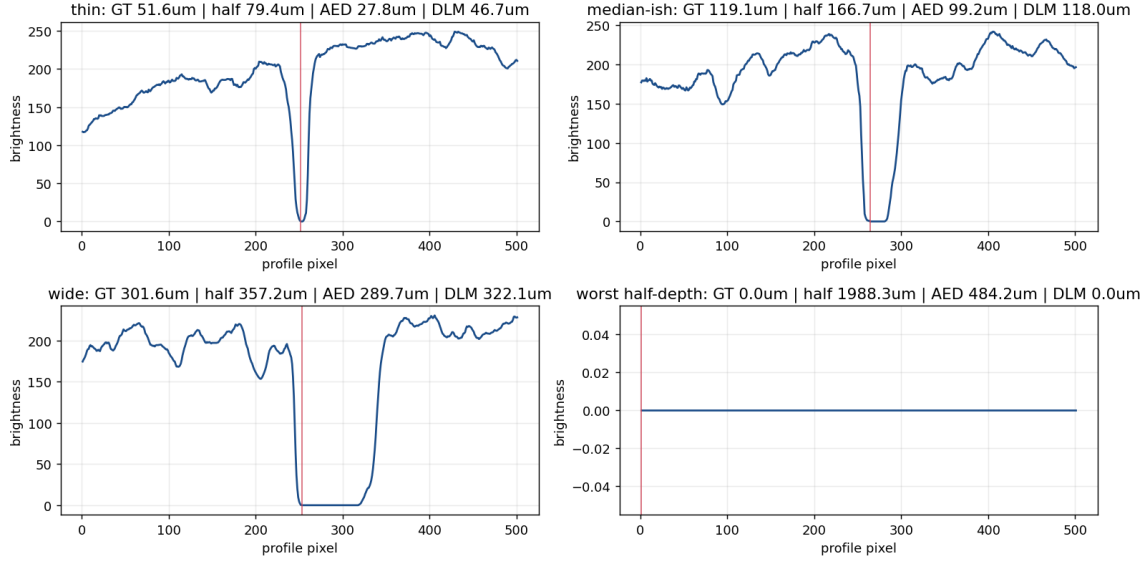


Figure 3: krkCMD cross-crack brightness profiles (501 samples, $3.97\mu\text{m}/\text{px}$) with manual width GT—the physical ground truth behind Table 2.

4.7 Boundary regimes: fuzzy and boundaryless defects

Binary segmentation collapses to chance on Magnetic-Tile fray/uneven ($\text{AUC} \approx 0.50$). With val/test protocol: illumination-residual modeling recovers uneven to 0.669 test AUC (DRAEM-lite: 0.636); for fray, a matting head trained on *directional* synthetic compositing achieves real-fray preservation IoU 0.949 vs. guided-filter 0.860—after an honest failure: the blob-synthesis v1 won $20\times$ synthetically yet transferred at 0.483. Synthesis realism, not architecture, was the fix.

4.8 Metrology rigor

Two silent measurement killers, quantified: camera tilt (50° : 19.3% $\rightarrow$ 0.7% via PlaneScale) and prompt synonym collapse (0.000 $\rightarrow$ 0.374/0.352 via the prompt-set ensemble). A 14-test metrology self-check (width $\pm 10\%$, scale $\pm 5\%$, e2e $\pm 15\%$ on synthetic GT) runs without any model and gates every release.

4.9 Honest negative: dense counting

Zero-shot rebar counting fails: best prompt MAE 13.1 (GT mean 22.5), acc@10% 5%, and six prompt variants all fail—a capability gap, not vocabulary (the same model detects 60 separated coins reliably). SAHI tiling recovers to MAE 7.4–8.9 but dense touching ends remain undercounted (e.g. GT 81 $\rightarrow$ 40). Supervised or density-based heads remain necessary in this regime; we report this as a boundary of current promptable counting.

5 Limitations

(1) *Localization coverage*: the signal-width chain currently measures 46–66% of stations on scanner imagery; the gate is honest but coverage is an open engineering task (ridge tracking, hysteresis

linking). (2) *Field captures*: our physical-unit evidence comes from CAD ground truth, known-dimension objects, and scanner profiles; an ArUco+caliper field protocol (released with a printable board and capture checklist) awaits real deployments. (3) *Single image, single depth*: plane scaling degrades on depth-elongated objects (worst 34.6% on T-LESS obj. 1); depth-aware measurement is future work. (4) *Transfer of the profile CNN* beyond 6400-dpi concrete scans is unvalidated. (5) Some task heads are trained on datasets with unstated licenses and are released for research use only, clearly marked.

6 Conclusion

GaugeAnything turns promptable segmentation into promptable *measurement*. The decisive move is a decomposition—masks for localization, raw signal for quantity—validated on physical ground truth at 23.2 μm median crack width and 2.5% industrial part dimensions, with every number audited, every negative reported, and every artifact released: code and benchmarks ([GitHub](#)), task heads ([HuggingFace](#)), and a live project page (falcons-eyes.github.io/GaugeAnything).

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